

Grade 6 Accelerated
Unit 5
Lesson 1 Homework

Name _____
Date _____
Period _____

Alvaro baked cookies for his class. The number of each type of cookie baked is shown.

Type of Cookie	Number of Cookies Baked
Sugar Cookies	18
Peanut Butter Cookies	30
Snickerdoodle Cookies	20
Chocolate Chip Cookies	18
Oatmeal Cookies	24
Total Cookies	110

For questions 1 through 5, use the table to describe the ratio of Alvaro cookies.

- What is the ratio of peanut butter cookies to chocolate chip cookies?
- What is the ratio of snickerdoodle cookies to oatmeal cookies?
- For every ____ sugar cookie(s) there are ____ chocolate chip cookie(s).
- There are 4 oatmeal cookies for every ____ chip cookie(s).
- Complete the statements below to describe the ratio of total cookies.

a. The ratio of

☐ peanut butter
☐ snickerdoodle

cookies to total cookies is 3:11.

b. The ratio of sugar cookies to total cookies is

☐ $\frac{2}{11}$.
☐ $\frac{9}{55}$.

The number of each type of cookie eaten is shown.

Type of Cookie	Number of Eaten Cookies
Sugar Cookies	15
Peanut Butter Cookies	15
Snickerdoodle Cookies	2
Chocolate Chip Cookies	18
Oatmeal Cookies	10
Total Cookies	60

For questions 6 through 10, use the table that describes the number of cookies eaten by the class **and** the table of baked cookies from the previous page.

6. What is the ratio of peanut butter cookies eaten to peanut butter cookies baked?
7. What is the ratio of sugar cookies eaten to sugar cookies baked?
8. For every ____ snickerdoodle cookie baked, only ____ snickerdoodle cookie(s) were eaten.
9. There was 1 chocolate chip cookie eaten for every ____ chocolate cookie(s) baked.
10. Complete the statements below to describe the ratio of cookies eaten to total cookies eaten.
 - a.

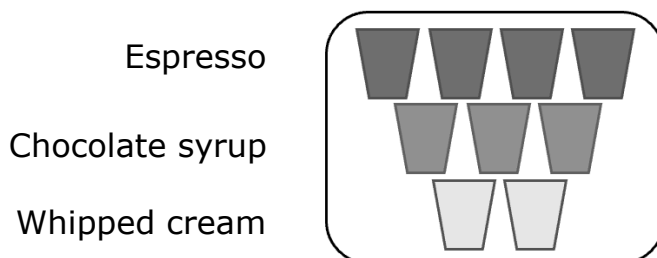
- ☐ Chocolate chip
 - ☐ Oatmeal

 cookies had a ratio of 1:6 of total cookies eaten.
 - b. For every ____ cookies eaten, ____ of the cookies were sugar cookies.

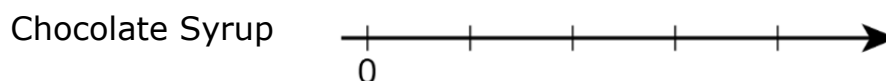
Grade 6 Accelerated
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Lesson 2 Homework

Name _____
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Jamal works at a gourmet coffee shop. Jamal always makes the Super Mocha Swirl when ordered. The representation shows the one-ounce portions of each ingredient.



For questions 1 through 5, use the double number line that represents amounts of espresso and chocolate syrup needed for Jamal to make Super Mocha Swirls.



- How many Super Mocha Swirls can be made with 16 ounces of espresso?
- How many ounces of chocolate syrup will be needed for 3 Super Mocha Swirls?
- For every ____ ounces of chocolate syrup in a Super Mocha Swirl, there are ____ ounces of espresso.
- Jamal noticed that he only has 24 ounces of chocolate syrup left. How many Super Mocha Swirls can be made before the next shipment of chocolate syrup?
- Jamal needs to place an order for

☐ 80
☐ 60

 ounces of espresso and

☐ 80
☐ 60

 ounces of chocolate syrup to have enough for the 20 Super Mocha Swirl pre-order.

Jamal also makes a special hot chocolate for the gourmet coffee shop. He uses a secret whipped cream to milk ratio of 1 cup to 1.5 cups.

Use the secret ratio to answer questions 6 through 10.

6. If Jamal has 8 cups of milk, how many cups of whipped cream will he need if he wants to use all 8 cups of milk for hot chocolate?
7. What is the equivalent ratio of whipped cream to milk for an order of 4 hot chocolates?
8. If Jamal is using 5 cups of whipped cream, _____ cups of milk will be needed for _____ hot chocolates.
9. For Jamal to meet his goal of 100 hot chocolates in a week, he will need _____ cups of whipped cream to _____ cups of milk.
10. Complete the statements below.
 - a. A ratio of

☐ 7:1.5

☐ 7:10.5

 is correct for 7 hot chocolates.
 - b. If 6 cups of whipped cream are mixed with

☐ 6

☐ 9

 cups of milk, _____ servings of hot chocolate are made.

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Lesson 3 Homework

Name _____
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Sam and Rohan's families are driving to Chicago, Illinois, from Orlando, Florida. Sam's family is driving a direct route of 1,157 miles at 65 mph. Rohan and his family are driving through Charlotte, North Carolina and will drive 1,250 miles at 70 mph.

1. At this rate, how many hours will it take Sam's family to get to Chicago, Illinois? (Round your answer to the tenths place.)
2. How many hours will it take Rohan's family to arrive in Chicago, Illinois? (Round your answer to the tenths place.)
3. Compare the two rates and complete the sentence.

According to these rates, the first to arrive in Chicago, Illinois, will be

- ☐ Sam
 - ☐ Rohan
 - ☐ Both
- .

Chelsee bakes cookies for the local hospitals. With her kitchen capacity she can bake 20 dozen cookies in 2 hours.

4. At this rate, how many cookies does she make in 1 hour?
5. If she works an 8-hour day, how many cookies will she make in one day?
6. How many days will she need to bake to produce 2,880 cookies?
7. Chelsee charges the hospitals \$400 for an 8-hour day. What is the unit rate of price per dozen?

Michael has been recording the number of field goals, touchdowns, and interceptions of the top 20 college football teams for the last month. His observations are represented in the table.

Type of Play	Number of Plays
Field goals	110
Touchdowns	200
Interceptions	75
Total games in the month	80

8. For every ____ field goals, ____ touchdowns were made.
9. What is the ratio of interceptions to total games in the month?
10. Complete the sentence.

The ratio of interceptions to touchdowns was

- ☐ 8:3
- ☐ 3:8

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Lesson 4 Homework

Name _____
Date _____
Period _____

Carrie is starting a dog-walking business. The table represents her first three jobs.

Miles of Walking	One Dog	Two Dogs	Three Dogs
1.5 miles	\$6		
2 miles			\$16
2.5 miles		\$15	

1. What does Carrie charge per one mile of dog-walking for one dog?
2. What is the unit rate for walking two dogs?
3. When Carrie walks three dogs, what does she charge per mile?
4. How much does the unit rate increase for each additional dog?
5. Carrie wants to purchase a pair of running shoes for \$75. She will need to walk _____ miles for the family with two dogs to purchase the shoes.
6. A family has hired Carrie to walk their three dogs every Tuesday and Thursday afternoon. If they agree to pay her \$36 every week, Carrie will walk the dogs

o 4

o 4.5

 miles each week.

Carrie has determined she needs a pricing sheet to be more efficient when quoting prices to dog owners.

Miles of Walking	One Dog	Two Dogs	Three Dogs
1 mile			
1.5 miles	\$6		
2 miles			\$16
2.5 miles		\$15	
3 miles			

7. Fill in the table by using the unit rates you determined previously.

Carrie has taken a partner to expand the dog-walking business. They have combined all money earned for one week to pay for business cards.

8. Carrie walked three dogs 4.5 miles, two dogs 3 miles, and one dog 5 miles during

the week. Carrie earned

- ☐ \$69
- ☐ \$74

to put towards business cards.

9. Her partner walked one dog 2.5 miles, two dogs 4 miles, and walked three dogs 2

miles. Carrie's partner earned

- ☐ \$50
- ☐ \$46

to put towards business cards.

10. Carrie will have enough money to purchase

- ☐ 500 glossy business cards for \$110.
- ☐ 1000 glossy business cards for \$150.

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Lesson 5 Homework

Name _____
Date _____
Period _____

Beverly has hired a new employee, Rosa, to help her make cupcakes for the local bakeries. She needs to determine if Rosa makes cupcakes faster, slower, or at the same speed than she does. The table represents the number of cupcakes baked hourly, in dozens.

Hours	Beverly's Cupcakes by the Dozen	Rosa's Cupcakes by the Dozen
1		
2	12	
3		12
4		
5		

Use the table for questions 1 through 10.

1. According to the table, Beverly bakes one dozen cupcakes at a rate of

☐ $\frac{12}{2}$.

☐ $\frac{12}{3}$.

2. Beverly bakes ____ dozen cupcakes in one hour.

3. What is Rosa's rate for 3 hours?

4. Rosa's unit rate indicates that she bakes ____ dozen cupcakes in ____ hour(s).

5. Rosa has a

☐ slower
☐ faster

unit rate than Beverly by ____ dozen(s).

6. Complete the table using the unit rates that you determined previously.

Beverly was curious as to how her rate of baking cupcakes compared to Rosa's rate throughout the day.

7. The ratio of the number of cupcakes Beverly baked to the number of cupcakes that

Rosa baked after 4 hours is

- ☐ 24 to 16.
- ☐ 20 to 20.

8. After an 8-hour day, Beverly baked _____ dozen cupcakes and Rosa baked _____ dozen cupcakes.

9. How did the unit rate change throughout the day?

10. Give some examples of why Beverly could have been slightly faster, or made slightly more cupcakes, than Rosa.

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Lesson 6 Homework

Name _____
Date _____
Period _____

A table representing gasoline prices per gallon at one gas station is shown.

Gas (gallons)	Price (dollars)
$\frac{1}{2}$	\$1.49
10	\$29.80
3	\$8.94
17	\$50.66

Use the table for questions 1 through 5.

- A car that holds ____ gallons of gas will pay \$59.60.
- A push mower typically uses 1 gallon of gas, which will cost ____ to fill up the tank.
- The road trip from Orlando, FL to Chicago, IL was 1152 miles. If the automobile got 24 miles per gallon on the trip,

☐ 42

 gallons of gas was needed at a cost of

☐ \$125.16

☐ \$143.04

.
- One unit, or gallon, of gas to its cost can be written as 1: ____.
- The gas station down the street has a unit rate of $\frac{1}{\$3.09}$. That station has an

☐ increased

☐ decreased

 unit rate than the station represented by the table.

The Shine Central Car Wash has an Ultimate Car Wash that uses 3.5L of soap and 2L of wax for every wash.

Soap	Wax	Car Washes
42L		12
	84L	
31.5L		

6. Complete the table that represents the amount of soap and wax used for the Ultimate Car Wash.
7. What is the unit rate of wax to every car wash?
8. The ratio of soap to wax for 12 car washes is ____:____.
9. The liters of soap needed for a day of

○ 53

○ 26

 Ultimate Car Washes is

○ 185.5

○ 52
10. A delivery of 460 liters of wax was just made to the Shine Central Car Wash. That is enough wax for ____ ultimate car washes. They will need

○ 805

○ 1104

 liters of soap to match that number of washes.

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Lesson 7 Homework

Name _____
Date _____
Period _____

1. Several rates are listed. Locate those that represent a unit rate. Select all that apply.
 - ☐ paying \$100 for an hour of car maintenance
 - ☐ driving 86 miles in 2 hours
 - ☐ biking 2 miles in 10 minutes
 - ☐ 2 cheeseburgers for \$1
 - ☐ \$25 per yard mowed
2. Jessica is shopping material for bedroom curtains. Her favorite material is on sale. The material is sold by the yard. Jessica set up a table to help determine the cost.

Material Needed	Cost
one panel – 6 yards	\$33.00
one top curtain – 3 yards	
tie backs -	\$5.50

- a. Complete the table for Jessica.
 - b. What is the unit rate for the material?
 - c. If Jessica spends \$126.50, she has purchased _____ yards of material for her curtains.
3. Complete the statement.

Unit rate for
☐ candies per dollar
☐ dollar per candies
 can be determined by
☐ multiplying
☐ dividing

 the amount of
☐ dollars
☐ candies
 by the amount of
☐ dollars
☐ candies
 .

4. If the unit rate of a Mini Bar candy bar is $\frac{1}{\$0.89}$, the cost of a case of 36 candy bars is _____.

Ice Treats employs 4 high school students. The table represents how much money each person earns in a workweek.

Employee	Weekly Pay	Hours Worked
Sarah	\$84.40	8
Micah	\$157.25	17
Stone	\$294	24
Haven	\$90	10

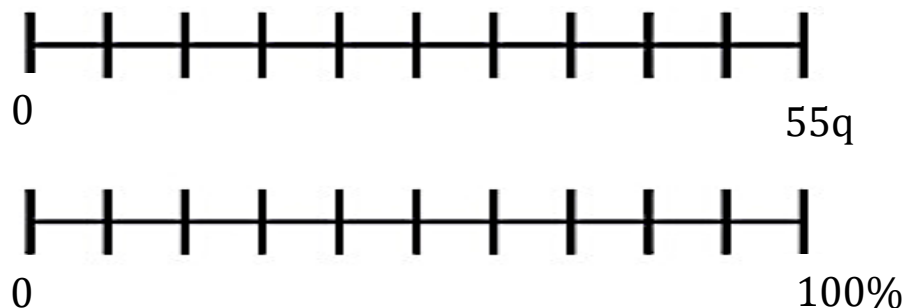
Use the table for questions 5 through 8.

5. What is the hourly rate for Stone?
6. Which employee earns a lower hourly rate, Sarah or Haven?
7. The manager is ☐ Micah ☐ Stone because he has the ☐ lowest ☐ highest hourly rate.
8. Sarah is on the schedule next week to work 12.5 hours. What will be her weekly pay?
9. One scoop of laundry detergent is 44.5 grams. That is a unit rate of ____ to _____. If the small box holds 1,230 grams, how many whole scoops are in the box?
10. The chocolate milk made for Maggie has a milk to chocolate syrup of $6c:3t$. What is the unit rate of milk to syrup?

Grade 6 Accelerated
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Lesson 8 Homework

Name _____
Date _____
Period _____

A large drink dispenser holds 55 quarts of sweet tea.



Use the number line diagram for questions 1 through 5.

1. Complete the diagram that represents the measurement and percentage of sweet tea in a dispenser.
2. How many quarts of sweet tea are in the dispenser if it is 40% full?
3. What percentage of the sweet tea is 38.5 quarts?
4. The dispenser will need to be refilled when the sweet tea has dropped to 20%,
or

☐ 10

☐ 11

 quarts.
5. If the dispenser is 80% full, how many quarts of sweet tea have been consumed?

The table represents the volume of a large water fountain and the corresponding percentages.

Volume	Percentage
1,340 gallons	100

6. As the maintenance person was filling the fountain, he wanted to keep a record of the gallons per 20% capacity. Complete the table.
7. If the water falls below 40%, the fountain pump will not work properly. What is the minimum gallons that must always be in the fountain?
8. The fountain is always circulating 268 gallons of water in the spraying system. What percentage of the water is always circulating in the spray?
9. When the fountain is at 80% capacity,

☐ 1072
☐ 804

 gallons of water are in the fountain.
10. The fountain loses 100 gallons a week due to evaporation. About how many weeks till the fountain is at 40%?

Grade 6 Accelerated
Unit 5
Lesson 9 Homework

Name _____
Date _____
Period _____

1. At a clothing store, a pair of jeans costs \$60. Using a coupon, Lyla buys the jeans for \$12 less than the original price. Complete the table.

Dollar Amount	Percentage

2. The coupon gives Lyla _____ % off of the original amount.
3. How much did Lyla pay for the jeans (before tax)?
4. In a bag of candy, 20% are blue. There are 30 candies in the bag. How many blue candies are there?
5. For one math test, Salvatore had to answer 24 questions. Of those 24 questions, he answered 22 correctly. What percent did Salvatore get on his math test? Round to the nearest percent.

6. Jeffrey has 40% as much money as Louis. Louis has \$80. Select all true statements.

☐ Jeffrey has \$32.

☐ Jeffrey has \$48.

☐ Jeffrey has \$24.

☐ Louis has $\frac{2}{5}$ as much as Jeffrey.

☐ Jeffrey has $\frac{2}{5}$ as much as Louis.

☐ Louis has $\frac{5}{2}$ as much as Jeffrey.

☐ Jeffrey has $\frac{5}{2}$ as much as Louis.

7. A small pool holds 20,400 gallons of water. Devyn filled it up to 85% capacity but then had to stop to do her other chores. Complete the table to determine how many gallons of water Devyn put in the pool.

Number of Gallons	Percentage of Capacity

8. Devyn will need to fill the pool with _____ more gallons of water to be at 100% capacity.

9. Many movie theaters offer a discount for students or children. At one local theater, a regular ticket for an evening show is \$17.99. A child's ticket for the same show is \$14.99. What percent of the full ticket price is a child's ticket? Round to the nearest percent.

10. Movie theaters often charge less for matinee showings. At the same theater, a matinee ticket for a child is 77% of the price of an adult's matinee ticket. Adult matinee tickets cost \$9.09. How much does a child's matinee ticket cost? Round to the nearest penny.

Grade 6 Accelerated
Unit 5
Lesson 10 Homework

Name _____
Date _____
Period _____

Use conversion rates to find each value. Show your work.

1. 3 days = _____ minutes

2. 5 liters = _____ milliliters

3. 60 cups = _____ gallons

4. 16 inches = _____ yards

5. 0.5 years = _____ days

6. 7.8 km = _____ cm

7. 2.25 quart = _____ pints

8. 6,000 lbs = _____ tons

9. A car traveling for 325 minutes has been traveling _____ hours.
10. Ephrosyni plays soccer at Polo Park Middle School in Wellington, Florida. She made a goal from 17 yards away. How many feet did she kick the ball to score the goal?

Grade 6 Accelerated
Unit 5
Lesson 11 Homework

Name _____
Date _____
Period _____

1. Each year, the ground crews at Walt Disney World log 450,000 miles to mow 2,000 acres of grass.

Use the table to determine how many miles ground crews log each year to mow the 727 acres of grass at Magic Kingdom. Show your work.

Grass to Mow (acres)	2000			
Distance Logged (miles)	450,000			

A shade of orange paint is created using a ratio of 5 quarts of yellow paint for every 6 quarts of red paint. Use this information to determine the amount of paint needed for different-sized batches of the same shade of orange paint in questions 2 through 4.

Yellow Paint (quarts)	Red Paint (quarts)
5	6
	3
12	
	1

A recipe for salad dressing uses 4 parts oil for every 3 parts vinegar. Use this information to answer questions 5 through 7.

5. Complete the table below to determine how much oil will be needed to make 28 teaspoons (tsp) of dressing.

Oil (tsp)	Vinegar (tsp)	Salad Dressing (tsp)

6. Complete the statement.

To make 28 tsp of salad dressing, ____ tsp of oil will be needed.

7. Determine an equivalent ratio for the recipe using 4.5 teaspoons of vinegar instead. Then, answer the question below.

How many teaspoons of salad dressing would the new combination make?

8. Addy went shopping with her mother and was trying to help her find the best deals. Addy noticed that the juice boxes she liked were on sale. She could get an 8 pack for \$3.59. To the nearest cent, what is the cost per juice box?

Addy next looked for chicken nuggets. She found two different brands with the prices shown. Use this information to answer questions 9 and 10.

Brand A	Brand B
\$6.99 for 12 ounces	\$8.09 for 16 ounces

9. Which brand of chicken nuggets is a better deal?
10. What is the price difference per ounce between Brand A and Brand B of chicken nuggets? Round your answer to the nearest cent.